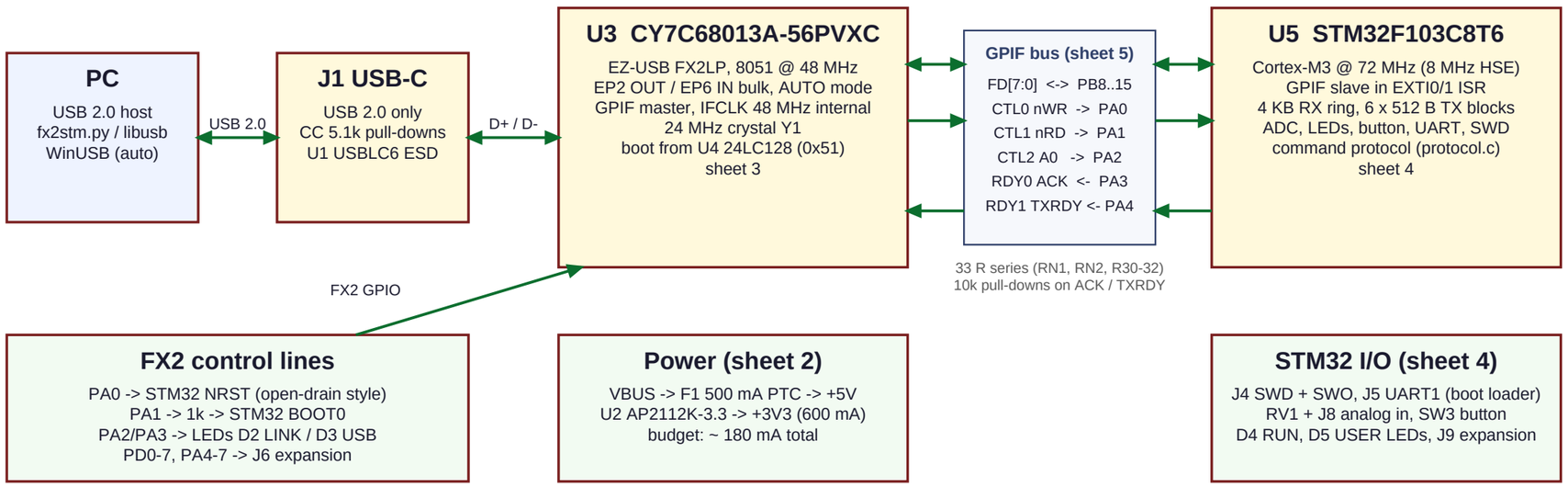
CY7C68013A-56 (EZ-USB FX2LP) <-> STM32F103C8T6 USB 2.0 / GPIF bridge

Rev A - FX2LP is USB high-speed device and GPIF master, STM32 is an asynchronous 8-bit bus slave



Sheet index

1 Overview / block diagram
2 USB-C connector, protection, power
3 CY7C68013A-56 FX2LP + boot EEPROM
4 STM32F103C8T6 + debug / user I/O
5 GPIF bus interconnect, timing, logic-analyser header

GPIF interface (8-bit, asynchronous 4-phase handshake)

Signal	FX2 pin	SSOP-56	QFN-56	STM32 pin	LQFP48	Dir	Function
FD0..FD3	PB0..PB3	25-28	18-21	PB8..PB11	45,46,21,22	bidir	data bit 0-3 (via RN1 33R)
FD4..FD7	PB4..PB7	29-32	22-25	PB12..PB15	25-28	bidir	data bit 4-7 (via RN2 33R)
nWR	CTL0	36	29	PA0	10	FX2->STM	write strobe, low = byte on FD valid (EXTI0 falling)
nRD	CTL1	37	30	PA1	11	FX2->STM	read strobe, low = STM32 drives FD (EXTI1 falling)
A0	CTL2	38	31	PA2	12	FX2->STM	1 = data stream, 0 = status register; nWR=nRD=0 -> link reset
ACK	RDY0	8	1	PA3	13	STM->FX2	handshake: STM32 raises after latch/drive, lowers after strobe ends
TXRDY	RDY1	9	2	PA4	14	STM->FX2	STM32 has >= 1 complete 512-byte block for the PC
NRST	PA0	40	33	NRST	7	FX2->STM	STM32 reset (FX2 drives low only, otherwise Hi-Z)
BOOT0	PA1	41	34	BOOT0	44	FX2->STM	via R16 1k; high + reset = STM32 ROM boot loader

Design notes

1. FX2LP package: CY7C68013A-56PVXC (SSOP-56) as drawn. For the 56-pin QFN (-56LTXC) QFN pin = SSOP pin - 7 (pin 1..7 -> 50..56); connect the exposed pad to GND.
2. USB D+/D-: 90 ohm differential, as short as possible, no stubs, no series resistors (FX2LP has internal terminations). Keep U1 next to J1.
3. Place RN1/RN2/R30-R32 close to U3. Keep the GPIF bus < 60 mm, solid GND plane below.
4. One 100 nF per VCC pin, as close to the pin as possible (C14-C19, C25-C28).
5. Y1: 24.000 MHz parallel resonant, 12 pF load, <= 500 uW drive (per FX2LP datasheet).
6. JP1 open = FX2 boots without EEPROM (enumerates 04B4:8613) - recovery / first flash.
7. STM32 BOOT0: JP2 or FX2 PA1. 10k pull-down keeps the STM32 in user flash by default.
8. All logic is 3.3 V. PB8..15 / PA0..4 of the STM32 are 5V-tolerant (FT).

Power budget

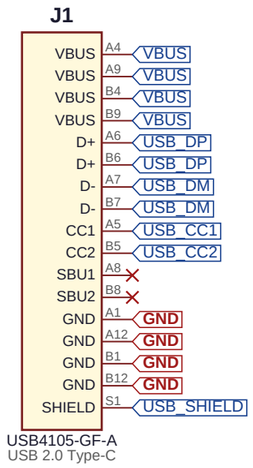
+3V3 load estimate (typ / max):

CY7C68013A @ 48 MHz, HS active	85 / 120 mA
STM32F103 @ 72 MHz, all periph.	36 / 50 mA
LEDs D1-D5 (2 mA each)	6 / 10 mA
24LC128, misc	1 / 3 mA
total	**~128 / 183 mA**

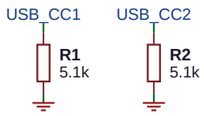
AP2112K-3.3: 600 mA, dropout 250 mV @ 600 mA
USB descriptor bMaxPower = 250 mA

PROJECT: **FX2-STM32 GPIF BRIDGE**	REV: **A**	DATE: 2026-10-02
SHEET TITLE: **Overview / block diagram**	SHEET: **1 of 5**	
DOCUMENT: FX2LP <-> STM32F103 GPIF bridge	SIZE: A3	UNITS: R: ohm C: F

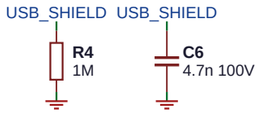
USB-C receptacle (USB 2.0)



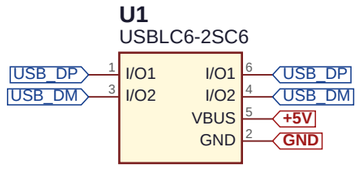
CC pull-downs (UFP, 5 V)



Shield to GND

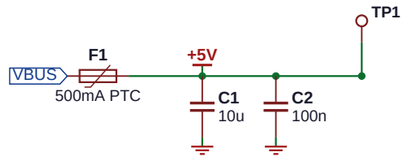


ESD protection

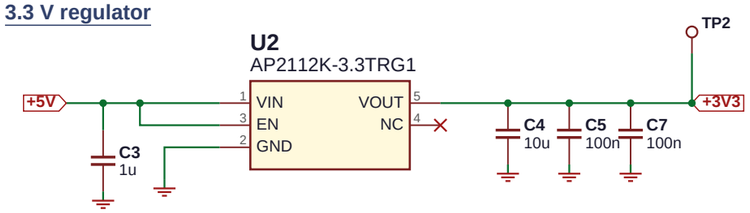


Pins 1/6 and 3/4 are flow-through pairs:
route D+/D- straight through under U1.

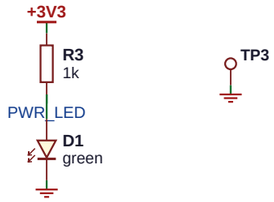
VBUS fuse and 5 V rail



3.3 V regulator



Power indicator



Notes

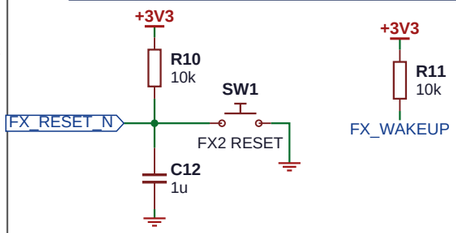
J1 is a USB 2.0-only Type-C receptacle: both D+ pins and both D- pins are joined; 5.1k on CC1/CC2 makes the board a 5 V / default-current sink.
USB-C to C and A to C cables both work.
F1 limits a short on +5V to the 500 mA USB budget.
AP2112K: Vin 5 V -> 3.3 V, EN tied to VIN, 1 uF in / 10 uF out ceramic.
Shield: 1M || 4.7 nF to GND (ESD path without a DC ground loop).

PROJECT: **FX2-STM32 GPIF BRIDGE**	REV: **A**	DATE: 2026-10-02
SHEET TITLE: **USB-C connector, protection, power**	SHEET: **2 of 5**	
DOCUMENT: FX2LP <-> STM32F103 GPIF bridge	SIZE: A3	UNITS: R: ohm C: F

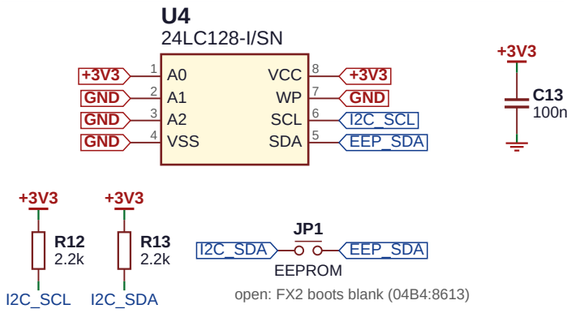
24 MHz crystal



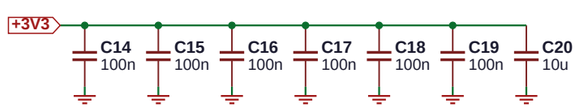
Reset (>= 5 ms after power-up) / wake-up



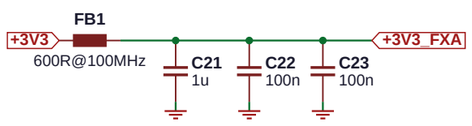
Boot EEPROM (I2C address 0x51 = FX2 'large EEPROM')



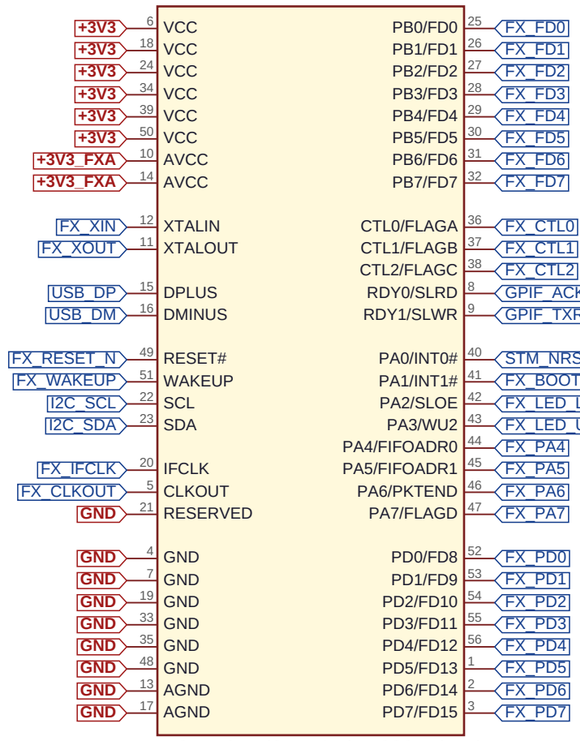
Decoupling (one 100 nF at every VCC pin)



Analog supply (AVCC) filter

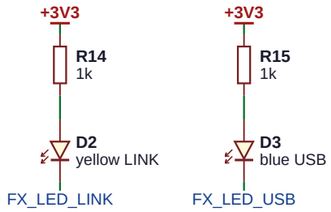


U3
CY7C68013A-56PVXC



EZ-USB FX2LP (QFN-56: pin = SSOP pin - 7)

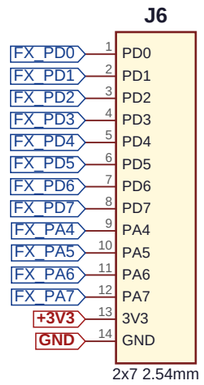
Status LEDs (active low)



STM32 BOOT0 drive



FX2 expansion header



Test points

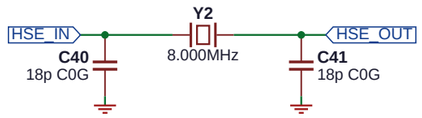


Notes

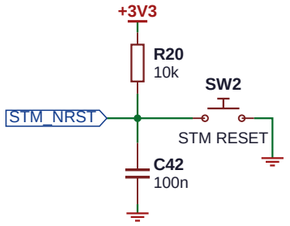
RESERVED (pin 21) must be tied to GND.
AGND (13, 17) joins the GND plane directly under U3.
PA0 drives STM32 NRST only low (OEA=1, IOA=0) or Hi-Z.
PA1 -> R16 -> BOOT0; JP2 on sheet 4 overrides it.
IFCLK is internal 48 MHz (not driven); CLKOUT disabled by firmware - both on test points for debugging.
I2C pull-ups are required even without EEPROM.

PROJECT: **FX2-STM32 GPIF BRIDGE**	REV: **A**	DATE: 2026-10-02
SHEET TITLE: **CY7C68013A-56 FX2LP + boot EEPROM**	SHEET: **3 of 5**	
DOCUMENT: FX2LP <-> STM32F103 GPIF bridge	SIZE: A3	UNITS: R: ohm C: F

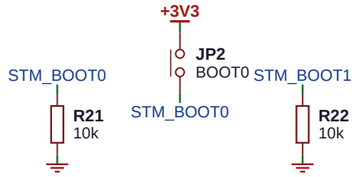
8 MHz HSE crystal



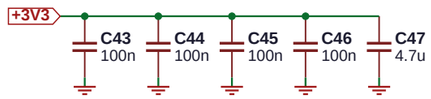
Reset



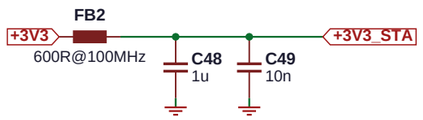
Boot mode



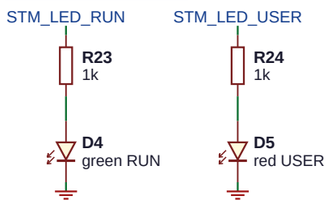
Decoupling



VDDA filter



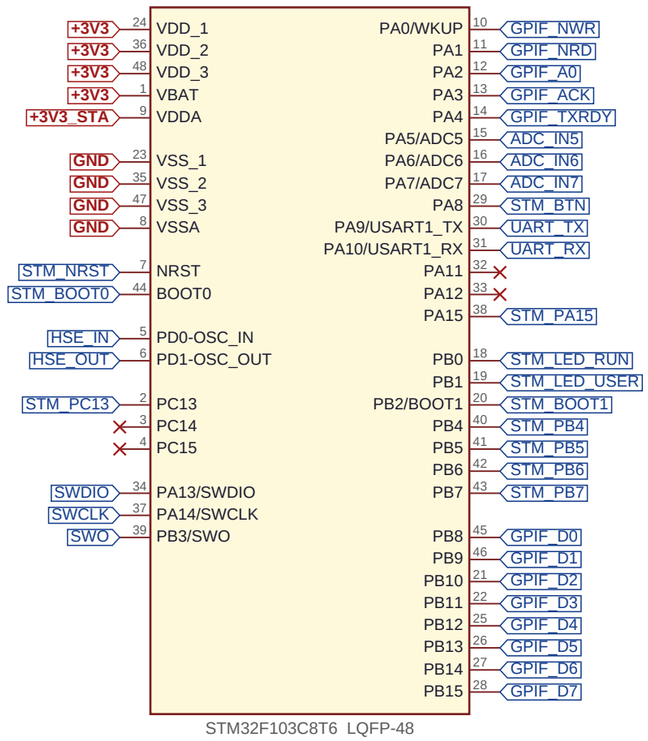
LEDs (active high)



User button / trimmer

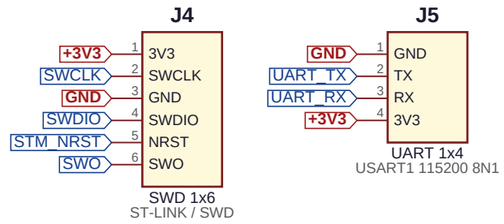


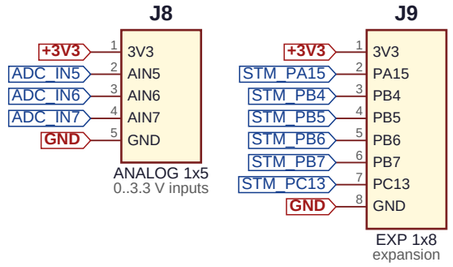
U5
STM32F103C8T6



STM32F103C8T6 LQFP-48

Debug and I/O headers



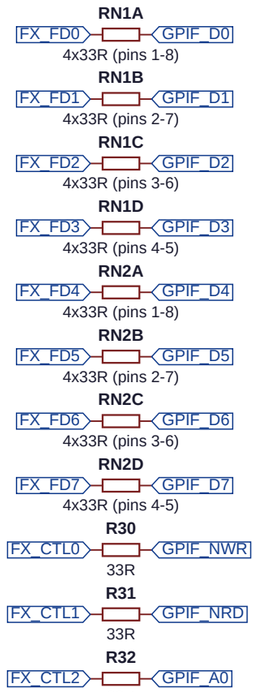


Notes

PB8..PB15 = GPIF_D0..D7 (see sheet 5), switched between input and push-pull output by GPIOB->CRH in the bus ISR.
PA0/PA1 strobes: EXTI0/EXTI1 falling edge, internal pull-ups.
PA3 ACK / PA4 TXRDY: push-pull, 10k pull-down (sheet 5) keeps them low while the STM32 is in reset.
JTAG is disabled (SWD only) -> PA15, PB4 free on J9.
USART1 (J5) is also the STM32 ROM boot loader port (BOOT0 = 1 via JP2 or FX2 PA1, then reset).
C40/C41: CL = 12 pF crystal -> 2 x (12 - 3) = 18 pF.

PROJECT **FX2-STM32 GPIF BRIDGE**	REV **A**	DATE 2026-10-02
SHEET TITLE **STM32F103C8T6 + debug / user I/O**	SHEET **4 of 5**	
DOCUMENT FX2LP <-> STM32F103 GPIF bridge	SIZE A3	UNITS R: ohm C: F

Series termination (place next to U3)

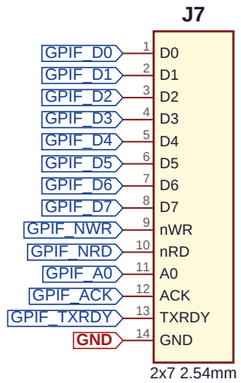


Handshake pull-downs

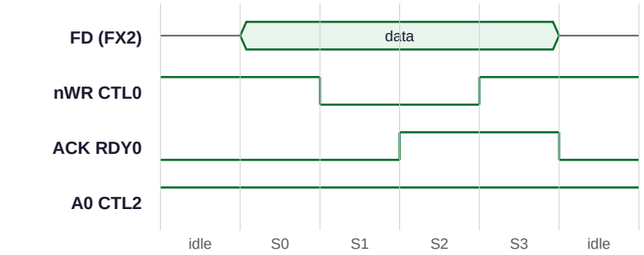


Hold RDY0/RDY1 low while the STM32 is in reset or unprogrammed.

Logic analyser header

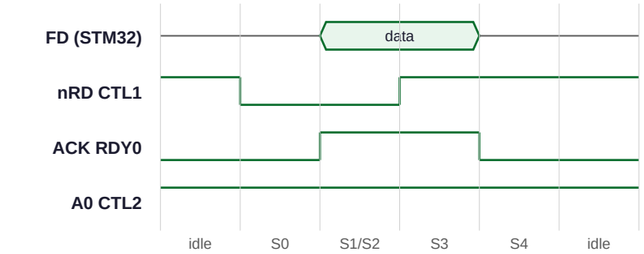


FIFO-WRITE (PC -> STM32), one byte per cycle, A0 = 1



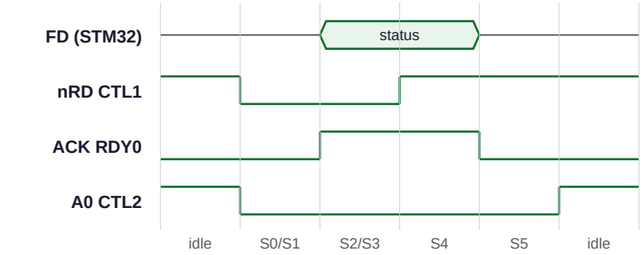
S0 data set-up (2 IFCLK). S1 nWR low, wait ACK=1 (STM32 latched FD).
S2 nWR high, wait ACK=0. S3 advance EP2 FIFO. ~0.2-0.4 us per byte.

FIFO-READ (STM32 -> PC), one byte per cycle, A0 = 1



nRD low; STM32 drives FD then raises ACK; FX2 samples FD into EP6 (S2);
nRD high; STM32 releases FD, then lowers ACK. Read only while TXRDY = 1.

STATUS READ (before every write burst), A0 = 0



status: bit7 = TXRDY, bit6..0 = free RX space / 32 bytes.
FX2 writes at most that many bytes -> the link cannot deadlock.
LINK RESET: nWR and nRD low together for ~100 us.

PROJECT: **FX2-STM32 GPIF BRIDGE**	REV: **A**	DATE: 2026-10-02
SHEET TITLE: **GPIF bus interconnect / timing**	SHEET: **5 of 5**	
DOCUMENT: FX2LP <-> STM32F103 GPIF bridge	SIZE: A3	UNITS: R: ohm C: F